Test of Mathematics for University Admission

DOOMSDAY

Set 1 · Paper 2

20 questions · 75 minutes · no calculator

A paper by

Viluxent (mkb6)

Twenty original questions, each calibrated to sit beyond the difficulty of an official Question 20. Every question lies inside the TMUA Paper 2 specification. None of them can be broken open by testing values, by working backwards from the options, or by recognising a familiar problem. Full worked solutions and an answer key follow the paper.

Before you begin

If you have come here from Paper 1, you already know what this series is. If you have not, sit Paper 1 first. This one assumes you have already been hurt once.

Paper 2 is a different animal. Paper 1 asked whether you could find the short route through a calculation. This paper asks whether you can tell the difference between an argument and something that merely looks like one. Most questions hand you statements, conditions or complete arguments, and ask you to decide exactly which of them survive scrutiny. The trap is almost never an arithmetic slip. It is a converse mistaken for the original, a quantifier read in the wrong order, a boundary case nobody checked, or a counterexample nobody bothered to look for.

The standard is the same as Paper 1: every question sits beyond an official Question 20. So 10 out of 20 is an outstanding score, and 5 or 6 on a first sitting is entirely normal. Sit it properly: under time, no calculator, no notes. Mark it honestly. Then, for every question you missed *or guessed*, write the argument out yourself from a blank page. On this paper more than any other, "I would have got that" does not count. If you cannot write down the counterexample or the proof, you did not have it.

A word on method. When a statement looks true, try to break it before you try to prove it; when it looks false, try to prove it before you go hunting for a counterexample. Almost every question here punishes whichever of those two instincts you have by default.

And, as always: if you get stuck on something in here, or on anything else (TMUA questions, revision plans, admissions interviews, what to read), **DM us**. We would much rather answer the question than watch you lose a week to it.

One last thing.

Consistency beats intensity.

Two focused hours every day will take you further than a ten-hour Sunday followed by five days of nothing. The candidates who reach **9.0** are rarely the ones with the most talent. They are the ones who kept turning up after the bad mock, after the question that made no sense, after the week in which nothing seemed to improve. The score is built on the days you did not feel like it.

Keep showing up.

See you all with those **9.0s**.

Viluxent (mkb6)

Candidate instructions

READ CAREFULLY

- You have **75 minutes** to attempt all **20** questions.
- **No calculator** may be used. No formula sheet is provided or needed.
- Each question has exactly one correct answer among the options given. **The number of options varies between questions**, so read each list carefully before answering.
- Where statements are labelled I, II and III, choose the option that lists **exactly** the statements asked for.
- There is no penalty for a wrong answer, so attempt everything.

A note on style

This is a Paper 2. Every question is about reasoning: whether a statement must hold, whether a condition is necessary, sufficient, both or neither, whether an argument is valid, and where a proof first goes wrong. Between them the twenty questions draw on logic and quantifiers, induction and contradiction, polynomials, functions and their graphs, sequences, integration, coordinate geometry, trigonometry, probability, number and counting. Everything needed is on the specification; nothing is needed that is not.

The options have been chosen adversarially. Where a question asks which statements are true, the false ones are statements that almost every candidate believes on first reading, and the true ones look too strong to be true. Testing a handful of values will usually confirm whatever you already believed, which is exactly why it does not work here.

Difficulty

This paper is aimed at candidates working towards **9.0**. Every question is calibrated to sit meaningfully beyond the difficulty of an official Question 20.

13–20	exceptional; you are ready
10–12	an outstanding result on this paper
6–9	strong; work through the solutions properly
0–5	entirely normal on a first sitting

Sources adapted for this set include the Korean CSAT, together with problems reworked from olympiad, invitational and entrance-examination material. No question is a reproduction: every one has been restructured, reframed as a question about reasoning, and re-verified so that the original answer is of no use.

Question 1

In each of the following, P denotes a polynomial whose coefficients are all integers.

Which of the following are possible?

I $\quad P(0) = 1, \quad P(2) = 3 \quad$ and $\quad P(4) = 9.$

II $\quad P(0) = 1, \quad P(2) = 5 \quad$ and $\quad P(4) = 25.$

III $\quad P\big(P(P(0))\big) = 0 \quad$ but $\quad P\big(P(0)\big) \neq 0.$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 2

Throughout, f is a function defined for all real numbers. The graph of $y = f(x)$ is *symmetric about the line* $x = a$ if $f(2a - x) = f(x)$ for all x, and *symmetric about the point* (b, c) if $f(2b - x) = 2c - f(x)$ for all x. A number $T > 0$ is a *period* of f if $f(x + T) = f(x)$ for all x, and f is *periodic* if it has a period.

Which one of the following statements is true?

A If the graph is symmetric about two different points, then there is a constant k such that $f(x) - kx$ is periodic.

B If the graph is symmetric about two different points, then f is periodic.

C If the graph is symmetric about the line $x = a$ and about the point (b, c), where $b \neq a$, then $2|b - a|$ is a period of f.

D If the graph is symmetric about the line $x = a$ and about the point (b, c), then the equation $f(x) = 0$ has at least one real solution.

E If the graph is symmetric about two different lines $x = a$ and $x = a'$, then $|a - a'|$ is a period of f.

F If the graph is symmetric about the line $x = a$ and about the point (b, c), then $f(a) = c$.

Question 3

The function f is defined for all real numbers and satisfies

$$f\big(f(x)\big) = -x \qquad \text{for every real } x.$$

(Such functions do exist.)

Which of the following statements must be true?

I $f(-x) = -f(x)$ for every real x.

II f is neither a strictly increasing function nor a strictly decreasing function.

III For every real $x \neq 0$, the numbers x and $f(x)$ have opposite signs.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 4

For each positive integer n, $S(n)$ is a statement. Three sets of conditions are proposed, each intended to guarantee that $S(n)$ is true for every positive integer n.

I $S(1)$ is true; for every $n \geq 1$, if $S(n)$ is true then $S(2n)$ is true; and for every $n \geq 4$, if $S(n)$ is true then $S(n-3)$ is true.

II $S(1)$ is true; for every $n \geq 1$, if $S(n)$ is true then $S(3n)$ is true; and for every $n \geq 3$, if $S(n)$ is true then $S(n-2)$ is true.

III $S(1)$ is true; and for every $n \geq 1$, if $S(1), S(2), \ldots, S(n)$ are all true, then $S(2n)$ is true.

Which of these sets of conditions guarantee that $S(n)$ is true for every positive integer n?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 5

For events X and Y with $\mathrm{P}(Y) > 0$, say that Y *supports* X if

$$\mathrm{P}(X \mid Y) > \mathrm{P}(X).$$

In the statements below, X, Y and Z are events, each with probability strictly between 0 and 1.

Which of the following statements must be true?

I If Y supports X and Z supports X, then $Y \cup Z$ supports X.

II If Y supports X, then X supports Y.

III If Y supports X and Z supports X, and Y and Z are mutually exclusive, then $Y \cup Z$ supports X.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

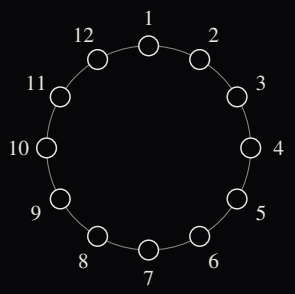
H I, II and III

Question 6

Twelve people sit around a round table, in seats numbered 1 to 12 in order, so that seat 12 is next to seat 1. Each person is either a *knight*, who always tells the truth, or a *knave*, who always lies. Every one of the twelve people says:

"Both of the people sitting next to me are knaves."

How many different assignments of knights and knaves to the twelve seats are consistent with these statements?



A 4

B 17

C 24

D 26

E 27

F 29

G 32

Question 7

A sequence of real numbers $a_1, a_2, a_3, \ldots$ is *convex* if

$$a_{n+1} \leq \tfrac{1}{2}\left(a_n + a_{n+2}\right) \qquad \text{for every } n \geq 1.$$

A two-way sequence $\ldots, b_{-2}, b_{-1}, b_0, b_1, b_2, \ldots$, with a term for every integer, is *convex* if $b_{n+1} \leq \frac{1}{2}(b_n + b_{n+2})$ for every integer n. A sequence is *bounded above* if there is a number M such that every term is at most M, and *bounded* if, in addition, there is a number m such that every term is at least m.

Which of the following statements are true?

I If a convex sequence $a_1, a_2, a_3, \ldots$ is bounded above, then $a_{n+1} \leq a_n$ for every $n \geq 1$.

II If a convex sequence $a_1, a_2, a_3, \ldots$ is bounded, then there is an N such that $a_n = a_N$ for every $n \geq N$.

III If a convex two-way sequence $\ldots, b_{-1}, b_0, b_1, \ldots$ is bounded above, then all of its terms are equal.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 8

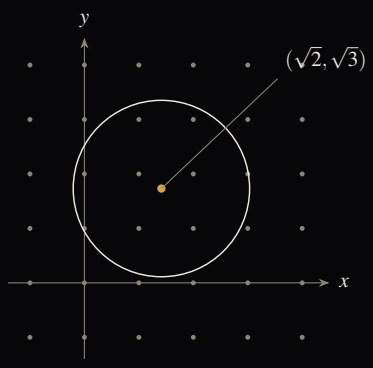
A *lattice point* is a point in the plane both of whose coordinates are integers.

Which of the following statements are true?

I No circle whose centre is $\left(\sqrt{2}, \sqrt{3}\right)$ passes through more than one lattice point.

II For every positive integer n, there is a circle which has exactly n lattice points strictly inside it.

III For every positive integer n, there is a circle whose centre is a lattice point and which has exactly n lattice points strictly inside it.



A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 9

What is the smallest positive integer n with the following property?

> Every set of n different integers chosen from $\{1, 2, 3, \ldots, 100\}$ contains three different numbers a, b and c such that a divides b and b divides c.

A 51

B 67

C 68

D 74

E 75

F 76

G 77

Question 10

Each of the conditions (1)–(5) below concerns a function f which is continuous on the stated interval and is not zero everywhere on it.

(1) On $[-1,1]$: $\displaystyle\int_{-1}^{1} f(x)\,\mathrm{d}x = 0$ and $\displaystyle\int_{-1}^{1} x\,f(x)\,\mathrm{d}x = 0.$

(2) On $[-1,1]$: $\displaystyle\int_{-1}^{1} f(x)\,\mathrm{d}x = 0$ and $\displaystyle\int_{-1}^{1} x^2 f(x)\,\mathrm{d}x = 0.$

(3) On $[0,1]$: $\displaystyle\int_{0}^{1} f(x)\,\mathrm{d}x = 0$ and $\displaystyle\int_{0}^{1} x^2 f(x)\,\mathrm{d}x = 0.$

(4) On $[-1,1]$: $\displaystyle\int_{-1}^{1} f(x)\,\mathrm{d}x = 0$ and $\displaystyle\int_{-1}^{1} x^3 f(x)\,\mathrm{d}x = 0.$

(5) On $[-1,1]$: $\displaystyle\int_{-1}^{1} f(x)\,\mathrm{d}x = 0$ and $\displaystyle\int_{-1}^{1} |x|\,f(x)\,\mathrm{d}x = 0.$

For how many of the five conditions is it true that *every* such function f has at least two zeros strictly inside the stated interval?

A 0

B 1

C 2

D 3

E 4

F 5

Question 11

A 7×7 board of unit squares has one of its 49 squares removed. The remaining 48 squares are to be covered exactly, with no overlaps, by sixteen 1×3 tiles, each tile covering three squares in a row or three squares in a column.

For how many of the 49 possible choices of the removed square can this be done?



tiles

A 1

B 4

C 5

D 9

E 13

F 16

G 17

H 25

Question 12

Consider the following claim.

> **Claim.** Let $n \geq 2$, and let $A_1, A_2, \ldots, A_n$ be arcs of a circle Γ, each arc being closed and not the whole of Γ. If every two of the arcs have at least one point in common, then some point of Γ lies on all n of the arcs.

A student offers the following argument.

(1) No arc A_i is the whole of Γ, so for each i there is a point of Γ that does not lie on A_i.

(2) Hence we may choose a point O of Γ that lies on none of the arcs $A_1, \ldots, A_n$.

(3) Cut Γ at O and straighten it into a line segment. Each arc becomes a closed interval $[l_i, r_i]$ of this segment, and two arcs have a common point exactly when the corresponding intervals do.

(4) Choose j so that l_j is the largest of $l_1, \ldots, l_n$, and k so that r_k is the smallest of $r_1, \ldots, r_n$.

(5) The intervals $[l_j, r_j]$ and $[l_k, r_k]$ have a common point, so $l_j \leq r_k$.

(6) Therefore $l_i \leq l_j \leq r_k \leq r_i$ for every i. So the point l_j lies in every interval, and the corresponding point of Γ lies on every arc.

Which one of the following is true?

A The claim is true, and the argument is a correct proof of it.

B The claim is true, but the argument is not a correct proof; its first error is in line (2).

C The claim is false, and the first error in the argument is in line (1).

D The claim is false, and the first error in the argument is in line (2).

E The claim is false, and the first error in the argument is in line (3).

F The claim is false, and the first error in the argument is in line (5).

G The claim is false, and the first error in the argument is in line (6).

Question 13

The following argument is intended to prove that there are infinitely many primes that leave remainder 3 on division by 4. The definition of N has been left blank.

> Suppose, for contradiction, that there are only finitely many such primes, and list them all as $p_1 = 3,\ p_2 = 7,\ \ldots,\ p_r$. Let $P = p_1 p_2 \cdots p_r$, and let $N =$ $\boxed{}$.

(1) N is odd, $N > 1$, and N leaves remainder 3 on division by 4.

(2) Every odd prime leaves remainder 1 or 3 on division by 4, and a product of integers which all leave remainder 1 on division by 4 also leaves remainder 1. Hence N has a prime factor q which leaves remainder 3 on division by 4.

(3) None of $p_1, p_2, \ldots, p_r$ divides N, so q is not in the list. This is a contradiction.

For which one of the following choices of N is every line of the argument correct?

A $N = 4P + 3$

B $N = P^2 + 2$

C $N = 4P + 7$

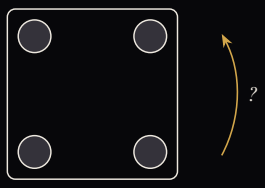
D $N = P + 2$

E $N = 2P + 3$

F $N = 6P - 1$

Question 14

Four coins lie at the corners of a square turntable, one at each corner, each showing heads or tails. You are blindfolded and do not know how the coins lie. A *move* consists of choosing two of the four coins, feeling them to find which way up each one is, and then turning over either, both or neither of those two coins. After every move the turntable is rotated through an unknown multiple of 90°, so you can never tell which coins you touched before. A bell rings as soon as all four coins show the same face.



The game is played under three different rules about which two coins may be chosen in a move.

I If any two of the coins may be chosen, you can guarantee that the bell rings within five moves.

II If the two coins chosen must always be at adjacent corners, you can guarantee that the bell eventually rings.

III If the two coins chosen must always be at opposite corners, you can guarantee that the bell eventually rings.

Which of these statements are true?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 15

A *word* is a string of letters, each of which is X or Y. A word is *safe* if every initial segment of it (the first letter, the first two letters, and so on) contains at least as many Xs as Ys. For a word w of length L and each $k = 0, 1, \ldots, L-1$, the *rotation* of w by k is the word obtained by moving the first k letters of w, in order, to the end. So a word of length L has L rotations, some of which may be the same word. For example, the rotations of XYY are XYY, YYX and YXY.



the safe word XXYXYY
(X up, Y down)

In the statements below, n is a positive integer.

I Every word with n Xs and n Ys has at least one rotation that is safe.

II For every word with n Xs and n Ys, exactly one of its $2n$ rotations is safe.

III For every word with $n+1$ Xs and n Ys, exactly one of its $2n+1$ rotations has the property that every initial segment contains strictly more Xs than Ys.

Which of these statements are true?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 16

For finite non-empty sets A and B of integers, write

$$A+B=\{a+b\,:\,a\in A,\,b\in B\},$$

and write $|S|$ for the number of elements of a set S. For example, if A is the set $\{0,1,3\}$, then $A+A$ is the set $\{0,1,2,3,4,6\}$, so that $|A+A|=6$. A set with only one or two elements counts as an arithmetic progression.

Which of the following statements are true?

I $|A+B|\geq|A|+|B|-1$ for all finite non-empty sets A and B of integers.

II If $|A+A|=2|A|-1$, then the elements of A, written in increasing order, form an arithmetic progression.

III If $|A|\geq 2$, $|B|\geq 2$ and $|A+B|=|A|+|B|-1$, then A and B are both arithmetic progressions, with the same common difference.

A none of them

B I only

C II only

D III only

E I and II only

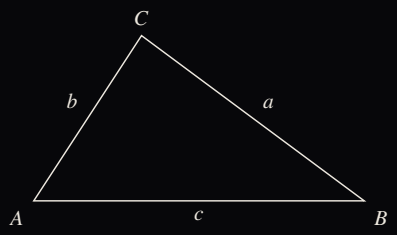
F I and III only

G II and III only

H I, II and III

Question 17

In triangle ABC the sides opposite the angles A, B and C have lengths a, b and c respectively.



Which one of the following statements is true for every triangle ABC?

A If $C < 60°$, then $a^2 + b^2 > 2c^2$.

B If $a^2 + b^2 > 2c^2$, then $C < 45°$.

C If $a^2 + b^2 > 5c^2$, then $c < a$ and $c < b$.

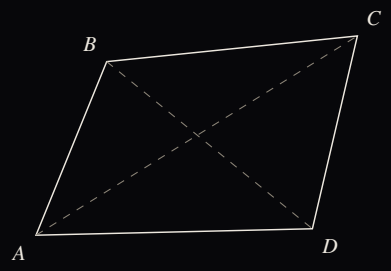
D If $a^2 + b^2 > 4c^2$, then $c < a$ and $c < b$.

E If $C < 60°$, then $c < a$ and $c < b$.

F If $a^2 + b^2 > 5c^2$, then $C < 30°$.

Question 18

$ABCD$ is a convex quadrilateral, with its vertices in that order. Say that one diagonal *bisects* the other if it passes through the midpoint of the other.



Which one of the following conditions, on its own, guarantees that $ABCD$ is a parallelogram?

A $\angle ABC = \angle CDA$, and the diagonal AC bisects the diagonal BD.

B $AB = CD$ and $\angle ABC = \angle CDA$.

C $\angle ABC = \angle CDA$, and the diagonal BD bisects the diagonal AC.

D $AB = CD$, and the diagonal BD bisects the diagonal AC.

E $AB = CD$, and AD is parallel to BC.

F $AC = BD$ and $\angle ABC = \angle CDA$.

Question 19

Claim. If x is a real number such that $x+\dfrac{1}{x}$ is an integer, then $x^n+\dfrac{1}{x^n}$ is an integer for every positive integer n.

Throughout, $t=x+\dfrac{1}{x}$ and $a_n=x^n+\dfrac{1}{x^n}$, so that $a_0=2$ and $a_1=t$. Three arguments are offered.

I For $n\geq 2$, multiplying out gives $a_{n+1}=t\,a_n-a_{n-1}$. Now $a_1=t$ is an integer. Suppose that $a_1,a_2,\ldots,a_n$ are all integers, for some $n\geq 2$. Then $a_{n+1}=t\,a_n-a_{n-1}$ is an integer. Hence, by induction, a_n is an integer for every $n\geq 1$.

II Since $x^2-tx+1=0$, the numbers x and $\dfrac{1}{x}$ are $\dfrac{t+s}{2}$ and $\dfrac{t-s}{2}$ in some order, where $s=\sqrt{t^2-4}$. So $a_n=\left(\dfrac{t+s}{2}\right)^n+\left(\dfrac{t-s}{2}\right)^n$. Expanding both brackets by the binomial theorem, the terms containing odd powers of s cancel, and every remaining term contains an even power of s, which is an integer power of the integer t^2-4. Hence a_n is an integer.

III For every $n\geq 1$, multiplying out gives $t\,a_n=a_{n+1}+a_{n-1}$, so $a_{n+1}=t\,a_n-a_{n-1}$. Now $a_0=2$ and $a_1=t$ are integers, and if a_{n-1} and a_n are integers for some $n\geq 1$, then so is a_{n+1}. Hence, by induction, a_n is an integer for every $n\geq 0$.

Which of the arguments are valid proofs of the claim?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 20

If a positive integer n has exactly k digits, its *echo* is the $2k$-digit number obtained by writing the digits of n twice in succession. For example, the echo of 2026 is 20262026.

Which of the following statements are true?

I If the echo of a k-digit number is a perfect square, then $10^k + 1$ is divisible by the square of some prime.

II There is a positive integer whose echo is a perfect square.

III If the echo of a k-digit number is a perfect square, then k is odd.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Answer key

Mark before reading on

Q	Ans	Value	Q	Ans	Value
1	**C**	II only	11	**D**	9
2	**A**	$f(x) - kx$ periodic	12	**D**	false; line (2)
3	**E**	I and II only	13	**B**	$P^2 + 2$
4	**A**	none of them	14	**B**	I only
5	**G**	II and III only	15	**F**	I and III only
6	**F**	29	16	**H**	I, II and III
7	**F**	I and III only	17	**C**	$a^2 + b^2 > 5c^2 \Rightarrow c$ shortest
8	**E**	I and II only	18	**C**	BD bisects AC
9	**F**	76	19	**D**	III only
10	**D**	3	20	**E**	I and II only

Mark honestly, then read the solution to every question you did not get out cleanly, including the ones you guessed correctly. On this paper a correct guess is worth nothing to you until you can write down the proof or the counterexample that justifies it.

Question 1 *answer C*

The weak test. For integers $u \neq v$, $u-v$ divides $P(u)-P(v)$, because $u^k - v^k = (u-v)(u^{k-1}+u^{k-2}v+\cdots+v^{k-1})$ for every k. Statement I passes this test completely: $2 \mid 3-1$, $4 \mid 9-1$ and $2 \mid 9-3$. So it cannot be the whole story.

The strong test. Write $P(x) = \sum c_k x^k$. Then

$$P(4) - 2P(2) + P(0) = \sum_k c_k \left(4^k - 2\cdot 2^k + 0^k\right).$$

The bracket is 0 for $k=0$ and for $k=1$, and equals $2^{k+1}\left(2^{k-1}-1\right)$ for $k \geq 2$, a multiple of 8. Hence $P(4)-2P(2)+P(0)$ is *always* a multiple of 8.

I. $9-6+1=4$: impossible. **II.** $25-10+1=16$ passes, and $P(x)=2x^2-2x+1$ gives $1, 5, 25$: possible.

III. Let $a = P(0)$ and $b = P(a)$, so $b \neq 0$ and $P(b) = 0$. If $a = 0$ then $b = P(0) = 0$; if $b = a$ then $0 = P(b) = P(a) = a$. So $0, a, b$ are distinct and P sends $0 \to a \to b \to 0$. By the weak test, $a \mid b-a$, $(b-a) \mid -b$ and $b \mid a$, so $|a| \leq |b-a| \leq |b| \leq |a|$ and all three are equal. Then $b = -a$ (as $b \neq a$), and $|b-a| = 2|a| \neq |a|$. Contradiction: impossible.

Trap. Option E is exactly what pairwise divisibility alone produces.

Question 2 *answer A*

A is true. If the centres are (b,c) and (b',c') then $b \neq b'$ (equal b would force $c = c'$). Replacing x by $2b-x$ in $f(2b'-x) = 2c' - f(x)$ gives

$$f(x+2d) = 2c' - f(2b-x) = f(x) + 2e, \qquad d = b'-b,\ \ e = c'-c.$$

With $k = e/d$, the function $g(x) = f(x) - kx$ satisfies $g(x+2d) = g(x)$, so $2|d|$ is a period of g.

The others. B: $f(x) = x$ is symmetric about every point of its graph and is not periodic. For a line $x = a$ and a point (b,c), the same substitution gives $f(x+2(b-a)) = 2c - f(x)$, so the guaranteed period is $4|b-a|$; C fails for $\sin x$ ($a = \frac{\pi}{2}$, point $(0,0)$), since $\sin(x+\pi) = -\sin x$. D fails for $2+\sin x$ (line $x = \frac{\pi}{2}$, point $(0,2)$), which is never 0. E fails for $\cos x$ (lines $x = 0$, $x = \pi$): the period is 2π, not π. F fails for $\sin x$ as in C, where $f(a) = 1 \neq 0$; what is true is $f(b) = c$.

Question 3 *answer E*

I is true. Applying f to both sides of $f(f(x)) = -x$ gives $f(f(f(x))) = f(-x)$. Applying the identity to the number $f(x)$ gives $f(f(f(x))) = -f(x)$. So $f(-x) = -f(x)$.

II is true. If f is strictly increasing, so is $f \circ f$. If f is strictly decreasing then $x < y \Rightarrow f(x) > f(y) \Rightarrow f(f(x)) < f(f(y))$, so again $f \circ f$ is strictly increasing. But $x \mapsto -x$ is strictly decreasing.

III is false. Put $f(0) = 0$ and, for $k = 0, 1, 2, \ldots$,

$$f(x) = x + 1 \text{ on } (2k, 2k+1], \qquad f(x) = -(x-1) \text{ on } (2k+1, 2k+2],$$

with $f(x) = -f(-x)$ for $x < 0$. If $x \in (2k, 2k+1]$ then $x + 1 \in (2k+1, 2k+2]$, so $f(f(x)) = -x$; if $x \in (2k+1, 2k+2]$ then $f(f(x)) = -f(x-1) = -x$; negative x follow by oddness. Here $f(\frac{1}{2}) = \frac{3}{2}$.

Trap. The picture of "rotation through 90°" suggests III. A real function cannot rotate; it has to link numbers in cycles $x \to f(x) \to -x \to -f(x) \to x$, and half of each cycle keeps its sign.

Question 4 *answer A*

Each set of conditions is satisfied by some statement $S(n)$ that is false for some n.

I. Take $S(n)$: "$3 \nmid n$". Then $S(1)$ holds, and neither doubling nor subtracting 3 changes whether a number is a multiple of 3. $S(3)$ is false.

II. Take $S(n)$: "n is odd". Tripling and subtracting 2 preserve oddness. $S(2)$ is false.

III. Take $S(n)$: "n is a power of 2". The hypothesis "$S(1), \ldots, S(n)$ all true" holds only for $n = 1, 2$, and then $2n \in \{2, 4\}$ is a power of 2. So the condition is satisfied, but $S(3)$ is false.

Trap. I and II look like genuine "go up, then come down" induction: the upward steps are unbounded and there is a downward step. What goes wrong is that the downward step moves in jumps of 3 or 2 and never leaves a residue class. III is strong induction with the conclusion $S(n+1)$ replaced by $S(2n)$, which never reaches an odd number.

Question 5 *answer G*

Multiplying by $\mathrm{P}(Y) > 0$, "Y supports X" means exactly $\mathrm{P}(X \cap Y) > \mathrm{P}(X)\mathrm{P}(Y)$.

II is true. That inequality is symmetric in X and Y.

III is true. If $Y \cap Z = \varnothing$, then

$$\mathrm{P}\big(X \cap (Y \cup Z)\big) = \mathrm{P}(X \cap Y) + \mathrm{P}(X \cap Z) > \mathrm{P}(X)\mathrm{P}(Y) + \mathrm{P}(X)\mathrm{P}(Z) = \mathrm{P}(X)\,\mathrm{P}(Y \cup Z).$$

I is false. Roll a fair die, with $X = \{1, 2\}$, $Y = \{1, 3\}$, $Z = \{1, 4\}$. Then $\mathrm{P}(X) = \frac{1}{3}$ and $\mathrm{P}(X \mid Y) = \mathrm{P}(X \mid Z) = \frac{1}{2}$, but $Y \cup Z = \{1, 3, 4\}$ gives $\mathrm{P}(X \mid Y \cup Z) = \frac{1}{3}$. The shared outcome 1 is what makes Y and Z each look good; the union counts it only once.

Question 6 *answer F*

Translate. A knight's statement is true, so both of a knight's neighbours are knaves. A knave's statement is false, so every knave has at least one knight next to it. Hence no two knights are adjacent, no three knaves are consecutive, and there is at least one knight. Going round the table, the gap from each knight to the next is therefore 2 or 3 seats.

Count. With s gaps of 2 and t gaps of 3, $2s+3t=12$, so $(s,t)=(6,0)$, $(3,2)$ or $(0,4)$, with $k=s+t$ knights. Mark one knight: its seat can be chosen in 12 ways and the clockwise sequence of gaps in $\binom{k}{t}$ ways. Each assignment is counted once for each of its k knights, so there are $12\binom{k}{t}/k$ assignments:

$$\tfrac{12}{6}\cdot 1+\tfrac{12}{5}\cdot\tbinom{5}{2}+\tfrac{12}{4}\cdot 1=2+24+3=29.$$

Traps. 24 is the five-knight family alone; 26 and 27 each lose one of the two small families; 17 is the answer for ten seats.

Question 7 *answer F*

Let $d_n=a_{n+1}-a_n$. Convexity says exactly that $d_{n+1}\geq d_n$: the differences never decrease.

I is true. If some $d_k>0$, then $d_n\geq d_k$ for all $n\geq k$, so $a_{k+m}\geq a_k+md_k$ and the sequence is not bounded above. So bounded above forces every $d_n\leq 0$.

II is false. $a_n=\frac{1}{n}$ is convex, since $a_n+a_{n+2}-2a_{n+1}=\dfrac{2}{n(n+1)(n+2)}>0$, and lies between 0 and 1, but is never constant. Its differences increase towards 0 without reaching it.

III is true. Now the differences are non-decreasing over all integers. A positive difference makes the sequence unbounded as $n\to+\infty$, as in I. If $d_k<0$, then $d_n\leq d_k$ for every $n\leq k$, so $b_{k-m}\geq b_k+m|d_k|$ for every $m\geq 1$, which is unbounded in the other direction. So every difference is 0.

Trap. III is the sequence form of "a convex function on the whole line that is bounded above is constant". II is the same statement moved to a sequence with only one end, where it fails.

Question 8 *answer E*

I is true. Suppose different lattice points (p,q) and (r,s) are equally far from $(\sqrt{2},\sqrt{3})$. Expanding the squared distances,

$$m=2\sqrt{2}\,u+2\sqrt{3}\,v,\qquad u=p-r,\ \ v=q-s,\ \ m=p^2+q^2-r^2-s^2,$$

all integers. If $v=0$ then $2\sqrt{2}\,u=m$ forces $u=0$, since $\sqrt{2}$ is irrational; likewise $u=0$ forces $v=0$. If both are non-zero, squaring $2\sqrt{2}\,u=m-2\sqrt{3}\,v$ gives $8u^2=m^2+12v^2-4\sqrt{3}\,mv$, so $m=0$; then $2u^2=3v^2$, which is impossible, because the power of 3 dividing the left side is even and on the right it is odd.

II is true. By I, the lattice points all lie at different distances from $(\sqrt{2},\sqrt{3})$. As the radius of a circle with this centre grows from 0, the number of lattice points inside rises by exactly one each time the radius passes one of these distances, so it takes every value $1,2,3,\ldots$

III is false. A quarter-turn about a lattice point O maps lattice points to lattice points and preserves

distance from O. So the lattice points inside such a circle, other than O itself, come in groups of four, and the count is always 1 more than a multiple of 4. It is never 2.

Question 9 *answer F*

At most 75 numbers avoid a triple. Split $\{1,\dots,100\}$ into 50 chains, one for each odd m: $C_m = \{m, 2m, 4m, \dots\} \cap [1, 100]$. Any three numbers from one chain form a triple $a \mid b \mid c$. A set with no triple therefore takes at most two numbers from each chain. For the 25 odd numbers $m \geq 51$ the chain is just $\{m\}$, so it takes at most one from each of those. In total that is at most $25 \cdot 2 + 25 \cdot 1 = 75$.

75 is possible. $\{26, 27, \dots, 100\}$ has 75 elements, and $a \mid b \mid c$ with $a < b < c$ would need $c \geq 4a \geq 104$.

So every 76-element set contains a triple, and 75 is not enough: $n = 76$.

Traps. 51 is the answer for pairs $a \mid b$. $\{34, \dots, 100\}$ also avoids triples, which suggests 68, but it is not the largest such set. 75 is the size of the largest set, not the smallest n that forces a triple.

Question 10 *answer D*

The lemma. Suppose g is strictly increasing on the interval, and $\int f = \int gf = 0$. Because $\int f = 0$ and f is not identically zero, f takes both signs, so it has a zero. If it had only one zero c, then f would keep one sign on each side of c, with opposite signs on the two sides. Then $\big(g(x) - g(c)\big) f(x)$ would have constant sign and would not be identically zero, so its integral would be non-zero. But

$$\int \big(g - g(c)\big) f = \int gf - g(c) \int f = 0.$$

So f has at least two zeros.

The count. In (1) $g = x$, in (3) $g = x^2$ on $[0, 1]$ and in (4) $g = x^3$ on $[-1, 1]$; all three are strictly increasing, so all three conditions work. For (2) and (5), take $f(x) = x$. Both x^3 and $x|x|$ are odd, so both integrals vanish, but f has only one zero. That makes three.

Trap. The same x^2 that works on $[0, 1]$ fails on $[-1, 1]$. What matters is whether the weight is monotonic on the interval, not which power of x it is.

Question 11 *answer D*

Necessity. Number rows and columns 0 to 6 and colour square (i, j) by the remainder of $i + j$ on division by 3. Every tile covers one square of each colour, so the 48 remaining squares must include 16 of each. On the full board colour 0 occurs $3 \cdot 3 + 2 \cdot 2 + 2 \cdot 2 = 17$ times and the others 16 times, so the removed square has $i + j \equiv 0$. The same argument with $i - j$ gives $i - j \equiv 0$. Together these give $i \equiv j \equiv 0 \pmod 3$, leaving the nine squares with both coordinates in $\{0, 3, 6\}$.

Sufficiency. By symmetry it is enough to check a corner, the middle of an edge and the centre. For $(0, 0)$ or $(0, 3)$, tile the rest of row 0 with two horizontal tiles and the remaining 6×7 block with vertical tiles. For the centre $(3, 3)$, the four blocks

$$\{0,1,2\} \times \{0,..,3\}, \quad \{0,..,3\} \times \{4,5,6\}, \quad \{4,5,6\} \times \{3,..,6\}, \quad \{3,..,6\} \times \{0,1,2\}$$

are 3×4 or 4×3 rectangles covering everything except $(3,3)$, and each is tiled by four tiles. Answer: 9.

Traps. One colouring alone leaves 17 candidates. 4 recalls the familiar 8×8 version; 1 is the 5×5 answer.

Question 12 *answer D*

The claim is false. Take three arcs of 240° each, from 0° to 240°, from 120° to 360°, and from 240° round to 120°. Any two share a 120° arc, but the three 120° gaps between them cover the circle, so no point lies on all three.

Line (1) is true; line (2) does not follow. Line (1) says that *for each* arc there is a point missing it. Line (2) needs a point that misses *every* arc. The order of the quantifiers has been swapped. In the counterexample the arcs cover the circle, so no such O exists.

Lines (3)–(6) are correct once O exists: they are exactly the proof that pairwise-meeting closed intervals of a line have a common point. So the argument does prove the claim when the arcs fail to cover the circle, and only then.

Trap. Option B: seeing the gap in line (2) but deciding that the claim must still be true, because its interval version is.

Question 13 *answer B*

Every p_i is odd, so P is odd and P^2 leaves remainder 1 on division by 4. P itself leaves remainder 1 or 3, depending on whether r is even or odd, and r is unknown.

$N = P^2 + 2$ **works.** N is odd, larger than 1, and leaves remainder $1 + 2 = 3$ on division by 4. Line (2) is correct in general: an odd number all of whose prime factors leave remainder 1 would itself leave remainder 1. Dividing N by any $p_i \geq 3$ leaves remainder 2, so line (3) is correct.

The others fail. For $4P + 3$ and $4P + 7$, line (3) fails, since $3 \mid 4P + 3$ and $7 \mid 4P + 7$. For $P + 2$, line (1) fails when r is odd; for instance, if the list were just 3, then $N = 5$. For $2P + 3$ and $6P - 1$, line (1) fails, because $2P$ and $6P$ leave remainder 2, so N leaves remainder 1.

Trap. $4P + 3$ is the "obvious" analogue of Euclid's $P + 1$, and it is wrong for the most basic reason: 3 is on the list.

Question 14 *answer B*

I is true. Here is a five-move strategy; "opposite" and "adjacent" refer to the two coins touched.

1. Opposite pair: turn both to heads.
2. Adjacent pair: one of them is a head from move 1; turn the other to heads. Now at least three coins are heads. If the bell has not rung, exactly one is tails.
3. Opposite pair: if one is tails, turn it over and the bell rings. Otherwise both are heads, and they are the two neighbours of the tail; turn one of them over. Now the coins are H, H, T, T in order round the table.

4. Adjacent pair: if they match, turn both over and the bell rings. Otherwise turn both over; the coins now alternate H, T, H, T.
5. Opposite pair: they match; turn both over and the bell rings.

II is false. The two coins you do not touch are also adjacent. If the coins are not all the same, some adjacent pair has its two *complementary* coins showing different faces: if every complementary pair matched, all four coins would match. If the rotation always hands you that pair, the untouched coins differ, and the bell cannot ring.

III is false. If the rotation always hands you the same two opposite coins, the other two are never touched. If they start different, the bell never rings.

Trap. II and III feel like mild restrictions. In fact a single unlucky rotation, repeated, freezes the coins you cannot reach.

Question 15 *answer F*

Write $X = +1$ and $Y = -1$, and let $s_0 = 0, s_1, \ldots, s_L$ be the running totals of a word. The word is safe exactly when every $s_j \geq 0$.

I is true. With n of each letter, $s_{2n} = 0$. Choose $k \in \{0, \ldots, 2n-1\}$ with s_k as small as possible. The running totals of the rotation by k are $s_{k+j} - s_k$ before the wrap and $s_j - s_k$ after it, and all of these are ≥ 0.

II is false. $XYXY$ is safe after rotating by 0 and by 2; so is $XYXXYY$, whose rotations $XYXXYY$ and $XXYYXY$ are different safe words.

III is true. Now $s_{2n+1} = 1$. Extend the totals periodically by $s_{m+2n+1} = s_m + 1$. The rotation by k has all totals positive exactly when $s_m > s_k$ for every m with $k < m \leq k + 2n + 1$.

- *Existence.* Let k be the *last* index in $\{0, \ldots, 2n\}$ at which the minimum of $s_0, \ldots, s_{2n}$ occurs. Later indices in the same period give larger totals, and indices in the next period give at least $s_k + 1$.
- *Uniqueness.* If $k < k'$ both worked, then $s_{k'} > s_k$ (from k), while $s_k + 1 = s_{k+2n+1} > s_{k'}$ (from k'), so $s_{k'} \leq s_k$ because these are integers. This is a contradiction.

Trap. II and III read like the same statement. The single extra X is exactly what makes the good rotation unique. (III shows that the number of safe words is $\frac{1}{2n+1}\binom{2n+1}{n}$, the Catalan number; none of that is needed here.)

Question 16 *answer H*

Write $A = \{a_1 < \cdots < a_m\}$ and $B = \{b_1 < \cdots < b_n\}$.

I is true. $a_1 + b_1 < a_1 + b_2 < \cdots < a_1 + b_n < a_2 + b_n < \cdots < a_m + b_n$ are $m + n - 1$ different elements of $A + B$.

III is true. More generally, walk from $(1,1)$ to (m,n) in the grid of index pairs, increasing i or j by one at each step. The corresponding sums $a_i + b_j$ strictly increase, giving $m + n - 1$ elements of $A + B$. If $|A + B| = m + n - 1$, every such walk lists all of $A + B$ in increasing order. The pair (i, j) is the $(i + j - 1)$th point of any walk through it, so $a_i + b_j$ is the $(i + j - 1)$th smallest element of $A + B$, and therefore depends

only on $i+j$. In particular $a_{i+1}+b_j = a_i+b_{j+1}$, that is,

$$a_{i+1}-a_i = b_{j+1}-b_j \qquad \text{for all } i<m,\ j<n.$$

Since $m,n \geq 2$, all consecutive differences in both sets equal one number d.

II is true. If $|A| = 1$ there is nothing to prove; otherwise take $B = A$ in III.

Trap. It is natural to expect some irregular pair of sets to squeeze into the equality case. The walk argument shows that the equality case is completely rigid. The only genuine exceptions, one-element sets, are excluded from III and harmless in II.

Question 17 *answer C*

C is true. Suppose $c \geq a$. By the triangle inequality $b < a+c \leq 2c$, so $a^2+b^2 < c^2+4c^2 = 5c^2$. Hence $a^2+b^2 > 5c^2$ forces $c < a$, and in the same way $c < b$.

D fails, so the constant 5 cannot be lowered to 4: take $a = c = 1$ and $b = 1.9$. Then $a^2+b^2 = 4.61 > 4c^2$, but $c = a$.

The rest fail. A is the converse of the true statement "$a^2+b^2 > 2c^2 \Rightarrow C < 60°$" (which holds because $\cos C > \frac{a^2+b^2}{4ab} \geq \frac{1}{2}$). With $a = 3$, $b = 8$ we get $C = 60°$ at $c = 7$, and taking $c = 6.9$ gives $C < 60°$ but $a^2+b^2 = 73 < 2c^2$. The same triangle defeats E: $C < 60°$ only prevents c from being the *longest* side. For B, $a = b = 1$, $c = 0.9$ gives $\cos C = 0.595 < \cos 45°$. For F, $a = b = 1$, $c^2 = 0.39$ gives $\cos C = 0.805 < \cos 30°$, since the bound obtainable is only $\cos C > \frac{4}{5}$.

Question 18 *answer C*

C is sufficient. Let M, the midpoint of AC, lie on BD. The half-turn about M swaps A and C and sends B to a point B' on line BD, on the opposite side of AC from B, which is the side containing D. A half-turn preserves angles, so $\angle AB'C = \angle CBA = \angle CDA$. Both B' and D therefore lie on the arc, on that side of AC, of points from which AC subtends this angle (angles in the same segment). The point M is the midpoint of the chord AC, so it lies inside that circle, and the ray from M through D meets the arc only once. B' is on the same ray, so $B' = D$. Hence M is the midpoint of BD too, and a quadrilateral whose diagonals bisect each other is a parallelogram.

Counterexamples.

- A and F: the kite $A(0,0)$, $B(1,2)$, $C(4,0)$, $D(1,-2)$ has AC bisecting BD, $AC = BD = 4$ and $\angle ABC = \angle CDA$.
- B: $A(-4,-3)$, $B(-1,6)$, $C(0,5)$, $D(3,-4)$ has $AB = CD = \sqrt{90}$ and $\cos\angle ABC = \cos\angle CDA = 1/\sqrt{5}$, but $BC = \sqrt{2} \neq \sqrt{50} = AD$. This is the ambiguous case: triangles ABC and CDA share AC, have $AB = CD$ and equal angles opposite AC, but are not congruent.
- D: $A(-2,-3)$, $B(-1,0)$, $C(2,3)$, $D(3,0)$ has the midpoint of AC at the origin, on BD, and $AB = CD = \sqrt{10}$.
- E: any isosceles trapezium.

Trap. A and C differ only in *which* diagonal bisects which. The kite destroys A; the arc argument proves C.

Question 19 *answer D*

I is not valid. The step is only claimed for $n \geq 2$, so it needs a_1 and a_2 as starting values. a_2 is never shown to be an integer. It is one ($a_2 = t^2 - 2$), but the argument never says so, and as written it establishes nothing beyond a_1.

II is not valid. After cancelling,

$$a_n = \frac{1}{2^{n-1}} \sum_{k \text{ even}} \binom{n}{k} t^{n-k} \left(t^2 - 4\right)^{k/2},$$

an integer *divided by* 2^{n-1}. Nothing in the argument shows that the numerator is divisible by 2^{n-1}. It is, because the claim is true, but that is precisely what needed proving.

III is valid. $t\,a_n = \left(x + \frac{1}{x}\right)\left(x^n + \frac{1}{x^n}\right) = a_{n+1} + a_{n-1}$ holds for every $n \geq 1$. There are two base cases, and the step passes from the pair (a_{n-1}, a_n) to a_{n+1}: a correct two-term induction.

Trap. I and III are almost word for word the same. The only differences are where the base cases and the range of the step begin, and that is the whole difference between a proof and a non-proof.

Question 20 *answer E*

The echo of a k-digit number n is $n \cdot 10^k + n = n\,(10^k + 1)$.

I is true. Suppose $n(10^k + 1)$ is a square but $10^k + 1$ is divisible by no square of a prime. Then each prime p dividing $10^k + 1$ divides it exactly once, so p must also divide n. Hence $10^k + 1$ divides n, which is impossible, since $0 < n < 10^k + 1$.

II is true. We need $10^k + 1$ to have a square factor, and the binomial theorem supplies one. Since $10 = 11 - 1$,

$$10^{11} = (11 - 1)^{11} = -1 + 11 \cdot 11 - \tbinom{11}{2} 11^2 + \cdots \equiv -1 \pmod{121},$$

because every later term contains 11^2. So $10^{11} + 1 = 121q$, with $q = 826\,446\,281$. For $4 \leq m \leq 10$ the number $n = qm^2$ has exactly 11 digits ($16q \geq 10^{10}$ and $100q < 10^{11}$), and its echo is $121q^2m^2 = (11qm)^2$. For example, $n = 13\,223\,140\,496$.

III is false. The same idea with $100 = 101 - 1$ gives $100^{101} \equiv -1 \pmod{101^2}$, that is, $101^2 \mid 10^{202} + 1$. Then $n = m^2\,(10^{202} + 1)/101^2$ with $32 \leq m \leq 100$ has exactly 202 digits, and its echo is a square. So k can be even.

Traps. Stopping at I gives B. Finding $k = 11$ and then assuming that only odd k can work, because $11 \mid 10^k + 1$ requires k odd, gives H. Once the mechanism is visible, 101 does for even k exactly what 11 did for odd k.

Provenance and syllabus notes

The Korean CSAT supplies the form of Questions 2, 3 and 7. The CSAT's *bogi* format, a set of claims about a function or a sequence judged together, is the closest thing in any national examination to the TMUA's I/II/III question. Questions 8, 9, 11, 12, 16 and 17 are rebuilt from olympiad and invitational material: Steinhaus's circles, divisor chains, the tromino colouring, Helly's theorem on the line, the small-doubling case of the Cauchy–Davenport inequality, and a classical triangle inequality. In each case the original was either a construction or a single answer. Here it has been turned into a question about which claims survive, with the neighbouring false claims kept as options.

On syllabus

Every question sits inside the TMUA Paper 2 specification. The following were checked explicitly.

- **No limits and no continuity arguments are examined.** Question 7 uses only "bounded", defined in the question. Question 10 needs only that a continuous function which is not zero everywhere, and does not change sign, has a non-zero integral.
- **No modular arithmetic is assumed.** Question 13 is phrased with remainders. The congruences in the solutions to Questions 11 and 20 are shorthand for arguments with remainders and the binomial theorem.
- **Probability is limited to the definition of conditional probability** (Question 5), and geometry to the angle properties of circles and the sine and cosine rules (Questions 17 and 18).
- **No named theorem is needed.** Helly, Cauchy–Davenport, Steinhaus and the cycle lemma are mentioned here for provenance only; every solution is self-contained.

On the answer options

Every option list was built after the solution, from the wrong answers the question actually produces. In the statement questions, each false statement is one that a strong candidate believes on first reading: a converse (Questions 5 and 17), a quantifier read in the wrong order (Question 12), a one-sided version of a two-sided fact (Question 7), or a pattern that stops one prime too early (Question 20). Each true statement is one that looks too strong to hold. Questions 4 and 16 deliberately have "none" and "all" as their answers. Candidates who have learned that these options are never correct will find that this is punished rather than rewarded.

Viluxent mkb6

Doomsday Set 1 · Paper 2 · more sets to follow